



Gabriele Rolo

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Date of birth: 26/06/2000 **Nationality:** Italian

PROFESSIONAL PROFILE

Professional profile

Bachelor's degree in Computer Engineering, currently pursuing a Master's degree in **Software Engineering** at **Politecnico di Milano** (expected December 2026), with a focus on Robotics, Artificial Intelligence and Complex Systems Management.

Master's thesis in **deep learning for medical image segmentation**.

Complemented by academic projects in robotics (ROS2, motion planning, sensor integration) and embedded flight software for a CubeSat mission.

Backed by four years of professional software engineering in the energy sector (Open Fiber, ENI), delivering large-scale, production-grade integration systems. Solid experience in complex system orchestration, high-volume data processing, secure communications and relational databases, built on the MuleSoft / Anypoint Platform ecosystem.

WORK EXPERIENCE

Engineering Ingegneria Informatica Spa

City: Roma | **Country:** Italy

[07/11/2022 - Current]

Software Development Advisor

1. Development of integration solutions using **MuleSoft**, following the **API-Led Connectivity** approach (System, Process, Experience layers).
2. **Design and maintenance** of synchronous and asynchronous integration flows (batch processes, scheduled jobs, high-volume data processing).
3. Collaboration with cross-functional teams to ensure alignment between business needs and technical solutions.

PROJECTS

[01/03/2026 - Current]

Master's Thesis: deep learning for medical image segmentation with uncertainty quantification

- Development of deep neural networks (nnU-Net v2) for semantic segmentation of anatomical structures in volumetric medical imaging.
- Focus on **model calibration and voxel-wise uncertainty quantification** (temperature scaling, deep ensembles, Expected Calibration Error) to make predictions reliable and interpretable.

Robotics Projects

- Designed and implemented **motion planning and feedback control algorithms** for autonomous mobile robots in simulation.
- Developed modular robotics software using **ROS/ROS2**, including sensor integration, node communication, and control-system architecture.
- Implemented **localization, mapping, and autonomous navigation** algorithms, validating system performance in simulated environments.

Complete mobile outdoor robotic system - multidisciplinary team project

- Contributed to a full mobile outdoor robot within a multidisciplinary team, from system architecture to hardware-software integration.
- Owned the **docking station and energy-management subsystem**: charging logic, power distribution and reliability considerations.

[15/03/2026 - Current]

6S CubeSat - Flight Software Engineer (PoliSpace)

- Flight software in **embedded C** for a CubeSat mission on a dual-OBC architecture (NanoMind A3200,

FreeRTOS).

- Implementation of the mission-mode management lifecycle and telecommand handling.
- Work on the **CSP networking stack** over CAN and UART between ground and subsystems.

Neuromorphic Computing & Spiking Neural Networks

- Developed and optimized **Spiking Neural Networks (SNNs)** using **LAVA**, exploring biologically inspired learning mechanisms such as STDP.
- Developed hardware-aware implementations using **MLIR/IRON** targeting **AMD AI Engine (AIE)** architectures.

EDUCATION & TRAINING

[12/09/2023 - Current]

COMPUTER SCIENCE AND ENGINEERING

Politecnico di Milano

City: Milano | **Country:** Italy | **Thesis:** Calibration and Uncertainty Visualisation in Medical Image Segmentation

[04/05/2026]

Mulesoft Platform Architect

Salesforce

Mulesoft Developer certificate

Salesforce

[30/09/2019 - 13/12/2022]

Bachelor's Degree in Computer Engineering

University of Cittadella di Catania

[14/09/2014 - 29/06/2019]

High School Diploma in Information Technology

IIS Galileo Ferraris

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING: B2 **READING:** B2 **WRITING:** B2

SPOKEN PRODUCTION: B2 **SPOKEN INTERACTION:** B2

SKILLS

Programming

Python | C | C++ | ROS 2 | PyTorch | SQL | JS / PHP / React Native | Java

Embedded

Embedded C | CSP over CAN / UART | RTOS

Tools

Git | Docker | Linux | distributed and event-driven systems

Enterprise

Mulesoft | Anypoint Platform | Dataweave

Autorizzo il trattamento dei miei dati personali presenti nel CV ai sensi dell'art. 13 d. lgs. 30 giugno 2003 n. 196 - "Codice in materia di protezione dei dati personali" e dell'art. 13 GDPR 679/16 - "Regolamento europeo sulla protezione dei dati personali".